

uBITX VFO

Web-Based VFO / BFO Controller for Experimenter HAMs

User Manual

Ver 5.4F

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Covers basic operation and on-air settings for the ESP32 + Si5351
web-controlled VFO/BFO interface for the uBitx HF transceiver.

The same program can also be used with any double or single conversion rig
by adjusting appropriate parameters in program.

This is extension of single clock controller by Mirko Pavleski and detailed at
<https://www.hackster.io/mircemk/the-ultimate-smartphone-vfo-esp32-si5351-wireless-control-bc1fcd>

Acknowledging the wonderful support of OM Ashok, VU2KYZ in testing and support.

Note: All screen shots in this manual are taken on an Android phone in Landscape mode. This program
also works in Portrait mode.

Attention: This program and its details are open sourced and provided as it is for experimenter
amateur radio homebrewer to enjoy the hobby. You can Modify / Upgrade and give it back to
the community. No specific support to be expected. Collaborative suggestions and corrections
are welcome (vu2spf at gmail dot com). TRY AT YOUR OWN RISK.

0. Hardware

The Hardware is evolving as the need develops. This program uses ESP32 connected to Si5351 as follows:

- **ESP32** development board (any variant)
- **Si5351** clock generator module (I2C interface)
- **Optional**: Analog signal for S-meter (0-3.3V input on GPIO32)

Pin Connections

Pin	Purpose	Name in this program
ESP32 GPIO21	Si5351 SDA	
ESP32 GPIO22	Si5351 SCL	
ESP32 GPIO36	S-meter input (optional) (MAX 3.3V)	SMETER_PIN
ESP32 GPIO32	PTT	PTT_PIN
ESP32 GPIO14	LPF – Tx C	TX_LPF_C
ESP32 GPIO13	LPF - Tx B	TX_LPF_B
ESP32 GPIO4	LPF - Tx A	TX_LPF_A
ESP32 GPIO27	Wifi Switch	wifi_switch
Si5351 CLK0	12 MHz (Audio modulator/demodulator)	BFO2 in this program
Si5351 CLK1	33/57 MHz (Second IF mixer)	BFO1 in this program
Si5351 CLK2	45-75 MHz (First IF mixer / VFO)	VFO in this program

There may be some changes and additions to above connections, as needed during further development.

1. Introduction

This manual describes how to operate the uBITX VFO 5.4F web controller, an ESP32-based variable frequency oscillator (VFO) and beat-frequency oscillator (BFO) controller for the uBitx HF transceiver. The entire interface runs in a web browser, no app installation is required on the phone, tablet, or computer you use to control it.

The controller drives three Si5351 clock outputs: CLK2 generates the main VFO injection frequency, CLK1 generates BFO1 (the second IF mixer), and CLK0 generates BFO2 (the audio demodulator carrier). It covers all five HF amateur bands supported by a stock uBitx, with both LSB and USB operation, dual VFOs (A/B) with split capability, RIT, ten memory channels each for VFO frequency, BFO1, and BFO2, and a built-in S-meter.

This manual covers day-to-day operation and on-air settings only. It assumes the controller hardware has already been connected as in section 0 above, and the firmware flashed; it does not cover circuit construction, wiring, or firmware compilation.

2. Connecting to the Controller

The controller starts in one of two WiFi modes, selected automatically at power-up by reading a physical switch on the board (connected to WiFi Switch pin - GPIO27 now). Check the serial console (if you have one connected) at startup to confirm which mode is active and which address to browse to, the exact address can vary by network.

2.1 Access Point Mode

In this mode the ESP32 creates its own WiFi network rather than joining your home network. Default network name (SSID): ubitx, password: 12345678. Connect your phone, tablet, or laptop directly to this network, then open a browser to the address shown on the serial console (typically 192.168.4.1 for an ESP32 access point). This mode is useful in the field, at a club station, or anywhere a home network isn't available.

2.2 Station (Home Network) Mode

In this mode the ESP32 instead joins an existing WiFi network whose name and password are set in the firmware source code before flashing. Once connected, open a browser to the IP address (set as 192.168.1.200) or printed on the serial console at startup, or check your router's list of connected devices. If the configured network can't be reached within about 20 seconds, the controller automatically falls back to Access Point mode, so you're never locked out even if you're away from that network.

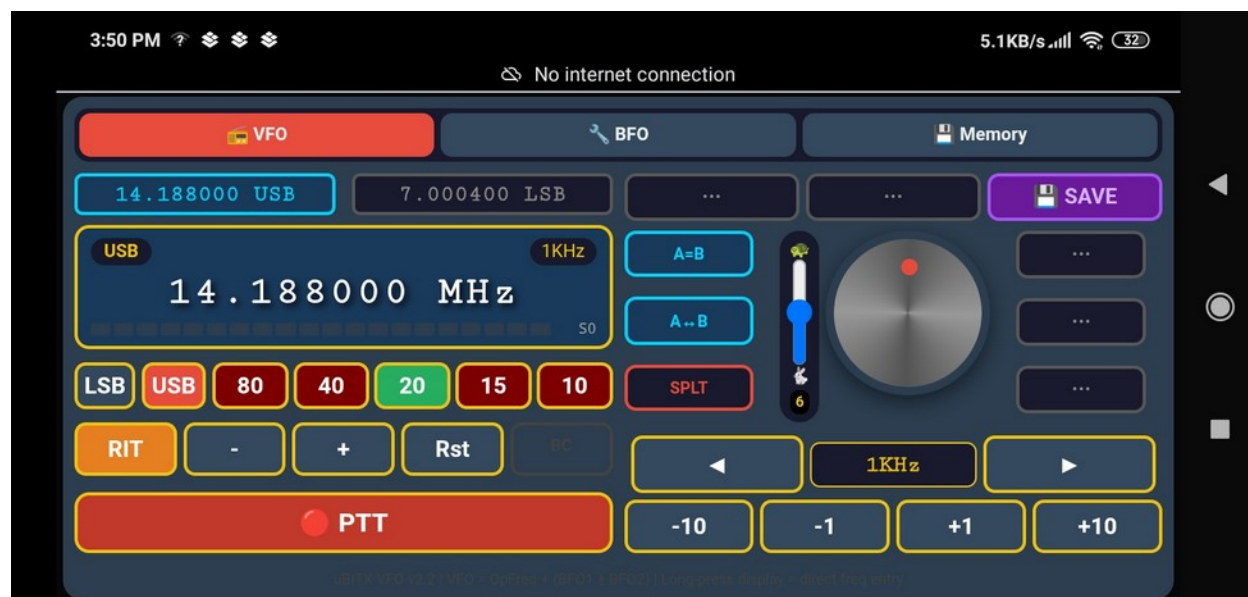
NOTE: If you are flashing this firmware onto your own hardware, open the source file and update the network name and password to match your own WiFi before building, then re-flash. Replace the “Your SSID” and “Password” around line 550.

2.3 Opening the Control Page

Once your device is connected to the right network, open any web browser and enter the IP address shown on the serial console. The interface loads as a single page with three tabs along the top: VFO, BFO, and Memory. Only one tab's content is shown at a time, but switching tabs does not interrupt reception, transmission, or anything else happening in the background.

3. The VFO Tab

This is the main operating screen and the one you'll use for almost everything. It opens by default whenever you load the control page.



VFO tab — main operating screen

3.1 Frequency Display and Dual VFOs

The large digits in the centre show the currently active operating frequency. Above the display, two labels show VFO A and VFO B, each with its own stored frequency and mode. Tap either label to make that VFO the one your knob, step controls, and band/mode buttons act on; the active one is highlighted. The mode and step size are shown at the top corners of the display.

3.2 Mode Selection (LSB / USB)

Tap LSB or USB to switch sideband immediately. Switching mode automatically retunes BFO1 to the calibrated point for that sideband (see Section 5). Each band also has a standard default mode, applied automatically whenever you switch bands, recall a memory, or type in a frequency directly:

Band	Frequency Range	Default Mode
80m	3.500 – 4.000 MHz	LSB
40m	7.000 – 7.200 MHz	LSB
20m	14.000 – 14.350 MHz	USB
15m	21.000 – 21.450 MHz	USB
10m	28.000 – 29.700 MHz	USB

3.3 Band Selection

The five band buttons (80, 40, 20, 15, 10) jump straight to the last frequency you used on that band, or a programmed default the first time you use it, and set that band's default mode automatically as shown in the table above.

3.4 Tuning Controls

- Tuning knob: drag with mouse or finger to change frequency.
- Sensitivity slider (turtle to rabbit icons): controls how much frequency change each knob movement produces, slide toward the rabbit for fast band-scanning, toward the turtle for fine tuning.
- Step display with  /  arrows: cycles the tuning step size through eight steps: 10 Hz, 100 Hz, 1 kHz, 5 kHz, 10 kHz, 100 kHz, 1 MHz, 10 MHz.
- Nudge buttons (-10, -1, +1, +10): move the frequency by that many steps with a single tap, useful for precise single-click adjustments.

3.5 RIT (Receiver Incremental Tuning)

RIT lets you fine-tune what you hear without moving your actual transmit frequency, useful when a station you're working is slightly off your transmit frequency. Tap RIT to turn it on or off. The + and - buttons step the offset by 100 Hz each press, up to ± 5 kHz. Rst clears the offset to zero and turns RIT off. While active, the RIT button shows the current offset.

3.6 Split Operation

- A=B: copies VFO A's current frequency and mode onto VFO B.
- A $\leftrightarrow$ B: swaps VFO A and VFO B (frequency and mode both swap together).
- SPLT: toggles split operation. While on, the radio receives on VFO A and transmits on VFO B, and a banner reading "SPLIT TX — transmitting on VFO B" appears on screen as a reminder.

3.7 Push-to-Talk and TX Safety

The PTT button toggles transmit on and off, tap once to transmit, tap again to return to receive. It is a toggle, not a press-and-hold control. Two safety behaviours are built in: if you leave the rig keyed for more than three minutes it automatically returns to receive, and if you're tuned outside the amateur bands (for example while listening to a time-standard station, see Section 3.9), the PTT button is disabled and a warning banner appears so you can't transmit out of band by mistake.

3.8 S-Meter

A 20-segment bar graph with an S-unit readout (S0 through S9+) shows received signal strength continuously while you're listening. Make sure that the analog input to SMETER pin is between 0 and 3.3v.

3.9 Time-Standard / Out-of-Band Indicator

The small "BC" indicator next to the RIT controls is a passive display, not a button. It shows the call sign of a known time-standard station (WWV or CHU) whenever you're tuned near one of

their broadcast frequencies, and changes appearance whenever your current frequency falls outside the amateur bands, a handy visual cue while band-scanning.

3.10 Quick-Save to Memory

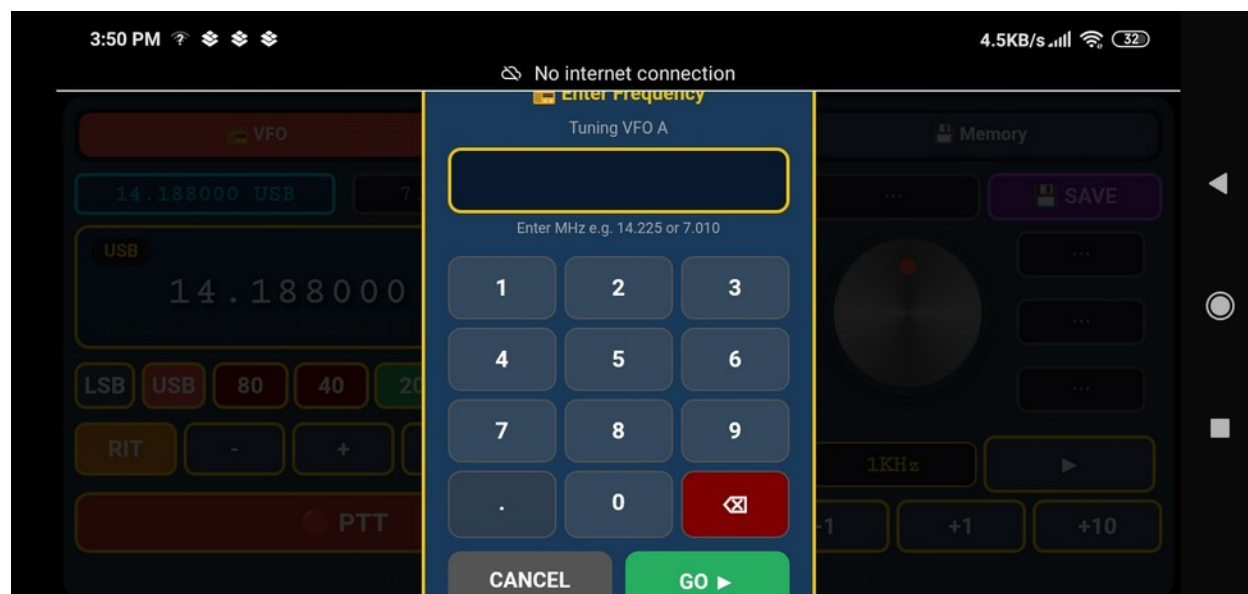
The SAVE button in the top-right of the VFO tab arms a 10-second save window and switches you to the Memory tab; tap any of the ten VFO memory slots within that window to store the current frequency and mode there. Full memory operation is covered in Section 7.

3.11 Inactive Buttons

There are 5 buttons near SAVE button which are for future development. (marked with "...").

4. Direct Frequency Entry

For jumping straight to a known frequency without spinning the knob, long-press (about half a second) anywhere on the main frequency display to open the entry screen.

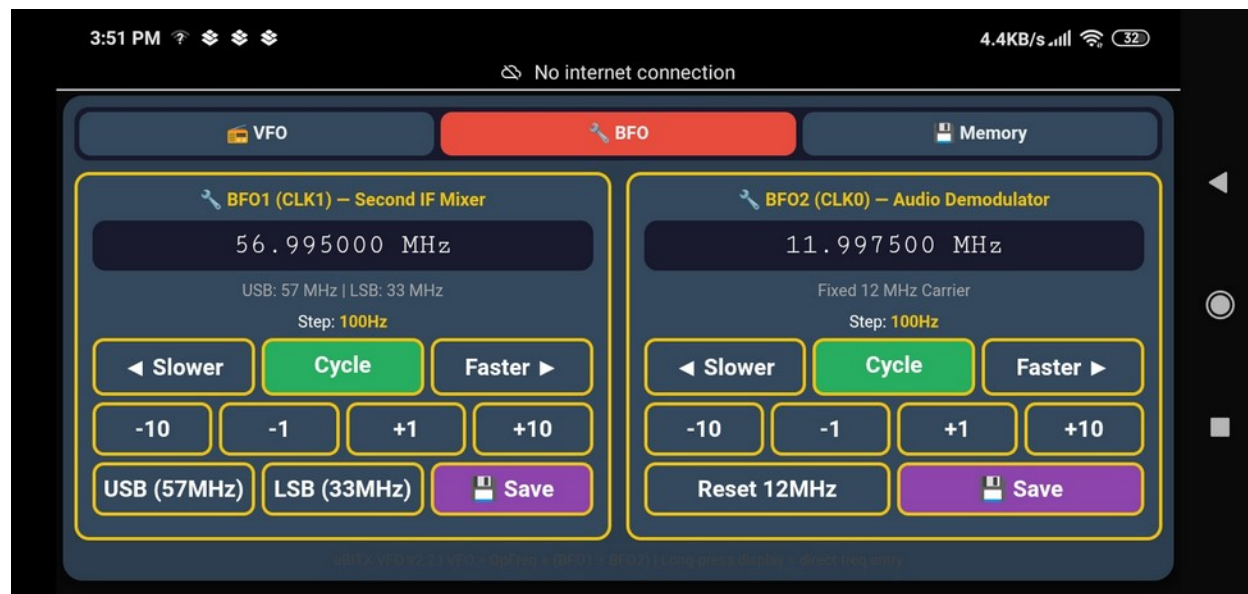


Direct Frequency Entry screen — numeric keypad overlay

- Type the frequency in MHz using the on-screen keypad, for example 14.225 for 14.225 MHz, or 7.01 for 7.010 MHz.
- Use the backspace key (⌫) to correct a mistake.
- Tap GO to apply the frequency, or CANCEL to close the screen without changing anything.
- The frequency must fall within one of the five amateur band ranges listed in Section 3.2; entries outside those ranges are rejected with an on-screen message rather than applied.
- The entry applies to whichever VFO (A or B) is currently active, and that band's default mode is applied automatically, exactly as with the band buttons.

5. The BFO Tab

This tab exposes the two oscillators that, together with the main VFO, determine what sideband you hear and transmit, and how cleanly the audio is demodulated. Most operators will rarely need this tab once the radio is calibrated; it exists for setup and the occasional touch-up, not for routine on-air use. The experimenter hams can play with these to learn their role and effect.



BFO tab — BFO1 / BFO2 setting screen

5.1 Understanding BFO1 and BFO2

BFO1 (CLK1) sets the radio's second mixing stage and is what actually selects upper versus lower sideband internally. It switches automatically between its calibrated USB point (around 57 MHz) and LSB point (around 33 MHz) every time the operating mode changes anywhere in the interface, from the Mode buttons, a band change, a memory recall, direct frequency entry, or a VFO swap. BFO2 (CLK0) is the fixed carrier, around 12 MHz, used in the final audio demodulation stage and normally does not need to change with mode. The relationship between all three oscillators is shown at the bottom of the page as a reminder: $\text{VFO} = \text{Operating Frequency} + (\text{BFO1} \pm \text{BFO2})$.

5.2 Adjusting BFO1 and BFO2

Each oscillator has its own step size, frequency readout, and adjustment controls. The Cycle button steps through seven step sizes: 10 Hz, 100 Hz, 1 kHz, 5 kHz, 10 kHz, 100 kHz, 1 MHz. The ◀ Slower and Faster ▶ buttons move one step size up or down the same list. The -10, -1, +1, and +10 buttons nudge the frequency by that many of the currently selected step, and the readout updates live as you adjust.

5.3 Presets and Saving

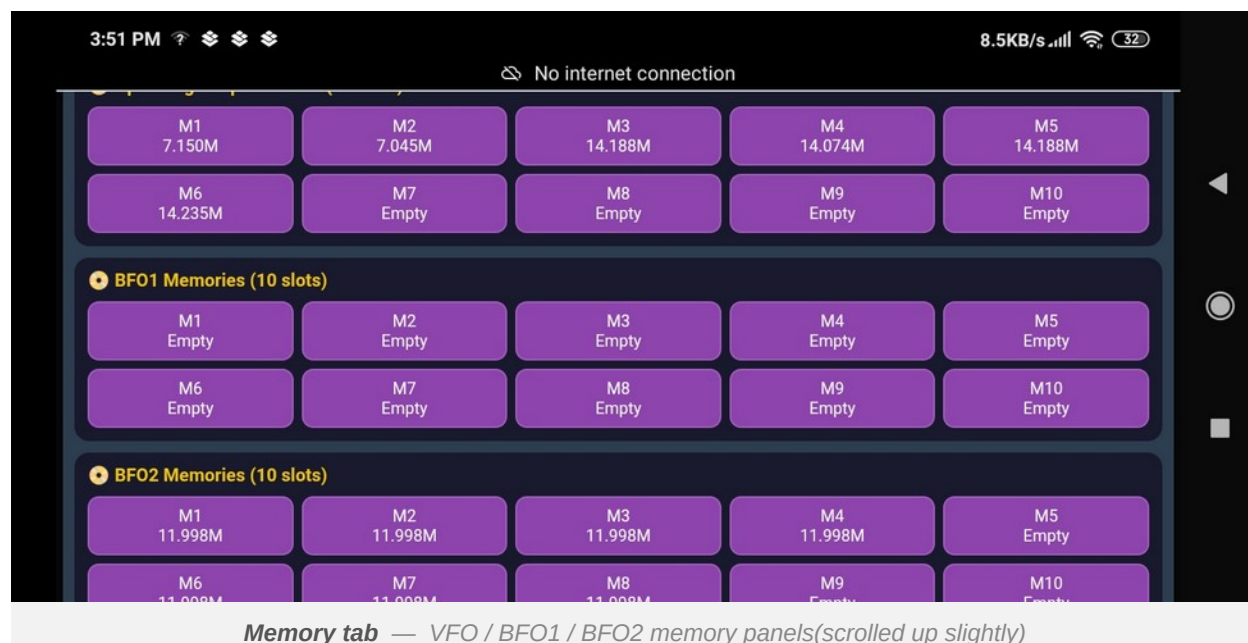
BFO1 has two one-tap presets, USB (~57 MHz) and LSB (~33 MHz), to quickly return to the factory default for either sideband after experimenting. BFO2 has a Reset 12MHz preset for the same purpose. Each oscillator also has its own Save button, which works the same way as the

VFO tab's SAVE button: it arms a 10-second save window and switches you to the Memory tab, where tapping a slot stores the current value there.

NOTE: Adjusting BFO1 or BFO2 changes how your receiver and transmitter sound, this is a calibration step meant to align the audio passband and sideband suppression with your specific crystal filter, normally done once on the bench, not something to adjust casually during a contact.

6. The Memory Tab

Three independent panels live on this tab: ten VFO memories, ten BFO1 memories, and ten BFO2 memories. Saving always starts from a SAVE button elsewhere in the interface (the VFO tab for VFO memories, the BFO tab for BFO1/BFO2 memories), this tab is where you complete that save, or simply recall something you stored earlier.



Memory tab — VFO / BFO1 / BFO2 memory panels(scrolled up slightly)

6.1 Saving

After tapping a SAVE button elsewhere, the interface switches to this tab and arms a 10-second window. Tap any memory slot in the relevant panel to store the current value there, overwriting whatever was previously in that slot, a confirmation message appears on screen. If you don't tap a slot within 10 seconds, the save is automatically cancelled and nothing is changed; just tap SAVE again to retry.

6.2 Recalling

Outside of a save window, tapping any filled memory slot immediately recalls that value, frequency and mode together for VFO memories, frequency alone for BFO memories, and the screen returns to the VFO or BFO tab a few seconds later. Empty slots are labelled "Empty" and tapping one does nothing but show a message.

7. Settings & Calibration Notes

- Autosave: changes you make are written to the controller's onboard memory automatically about 20 seconds after you stop adjusting something, so your last frequency, mode, and BFO calibration are restored after a power cycle. Memory slot saves, by contrast, are written immediately when you tap a slot, there's no waiting.
- BFO1 and mode are linked: as covered in Section 5.1, BFO1 always snaps to its USB or LSB calibration point the moment mode changes anywhere in the interface. If you've manually trimmed BFO1 while operating in one sideband, that trim is tied to that sideband and will be restored automatically whenever you return to it; the other sideband keeps its own separately calibrated value.
- First-time defaults: if a band has never been tuned before, the controller initializes it to a sensible default frequency the first time it boots, listed in Appendix A.

8. Troubleshooting

Symptom	What to check
Can't find the WiFi network, or can't connect	Check the position of the WiFi mode switch on the board (Section 2), and confirm whether the controller is in Access Point or Station mode. The serial console, if connected, shows the exact SSID and IP address in use.
Page loads but a control doesn't seem to respond	Reload the page. Switching away from a tab and back, or a full reload, pulls the controller's actual current state into the browser.
PTT button is greyed out and won't transmit	You're tuned outside the amateur bands. Retune within a band edge (Section 3.2) and PTT will re-enable automatically.
Transmit stopped on its own mid-contact	The three-minute safety timeout triggered (Section 3.7). Tap PTT again to re-key.
Wrong sideband heard despite the correct mode shown on screen	Check BFO1/BFO2 calibration on the BFO tab against your crystal filter (Section 5.3).
BFO1 doesn't change when running a split contact with VFO A and VFO B on different sidebands	Confirm you're running 5.4F or later; earlier versions could leave BFO1 tuned to VFO A's sideband during split transmit instead of VFO B's (see Appendix B).

Appendix A: Quick Reference

Band Plan

Band	Frequency Range	Default Mode
80m	3.500 – 4.000 MHz	LSB
40m	7.000 – 7.200 MHz	LSB
20m	14.000 – 14.350 MHz	USB
15m	21.000 – 21.450 MHz	USB
10m	28.000 – 29.700 MHz	USB

Other Defaults

- Tuning steps: 10 Hz, 100 Hz, 1 kHz, 5 kHz, 10 kHz, 100 kHz, 1 MHz, 10 MHz
- BFO steps: 10 Hz, 100 Hz, 1 kHz, 5 kHz, 10 kHz, 100 kHz, 1 MHz
- RIT range: ± 5 kHz, in 100 Hz steps
- PTT auto-timeout: 3 minutes
- Memory slots: 10 each for VFO, BFO1, and BFO2
- Default Access Point credentials: SSID ubitx, password 12345678
- BFO1 defaults: ≈ 57 MHz (USB), ≈ 33 MHz (LSB) — BFO2 default: ≈ 12 MHz

Appendix B: Version History

This section is here for other builders comparing firmware versions or deciding whether to update; it is not needed for day-to-day operation.

Version	Notable changes
5.4F	Mode-change requests (Mode button, band switch, memory recall, VFO swap, direct frequency entry) no longer get silently dropped by the tuning knob's network throttle. BFO1 is now correctly retuned to VFO B's sideband during split transmit when VFO A and B are set to different modes. The on-screen BFO1 reading now updates immediately after any automatic mode change instead of requiring a page reload.
5.3E and earlier	Corrected VFO/BFO relationship with working band selection. Mode changes sent shortly after a knob movement, band switch, or memory recall could be dropped without warning, leaving the radio's actual sideband out of sync with what the screen showed. Split transmit with VFO A and B on different sidebands could leave BFO1 tuned to the wrong one.